

Broadcast Campaign Error Analysis & Resolution Report

Project: Glowix Cosmetics / BuildStart

Component: WhatsApp Promotional Broadcast (BroadcastManager.tsx & broadcast-manager-Glowix_cosmetics)

Error Signature

Edge Function returned a non-2xx status code (HTTP 500)

Underlying PostgREST Error:

```
{
  "error": "Could not find the 'batch_cooldown_seconds' column of
'broadcast_campaigns' in the schema cache"
}
```

1. Executive Summary

When attempting to launch a WhatsApp promotional broadcast campaign from the BuildStart settings dashboard (/dashboard/settings), clicking the **"Launch Broadcast to X Contacts"** button triggered a red toast notification: **"Failed to Start: Edge Function returned a non-2xx status code"**.

Inspection of the Firefox Network tab revealed that the POST request to `https://supabase.buildstart.io/functions/v1/broadcast-manager-Glowix_cosmetics` failed with an **HTTP 500 Internal Server Error**. The response body confirmed the database constraint violation missing a specified column.

Core Root Cause: A schema synchronization mismatch exists between the newly developed Edge Function code and the live PostgreSQL database table `Glowix_cosmetics.broadcast_campaigns`. The Edge Function expects newer batch-delay management columns that do not currently exist in the database.

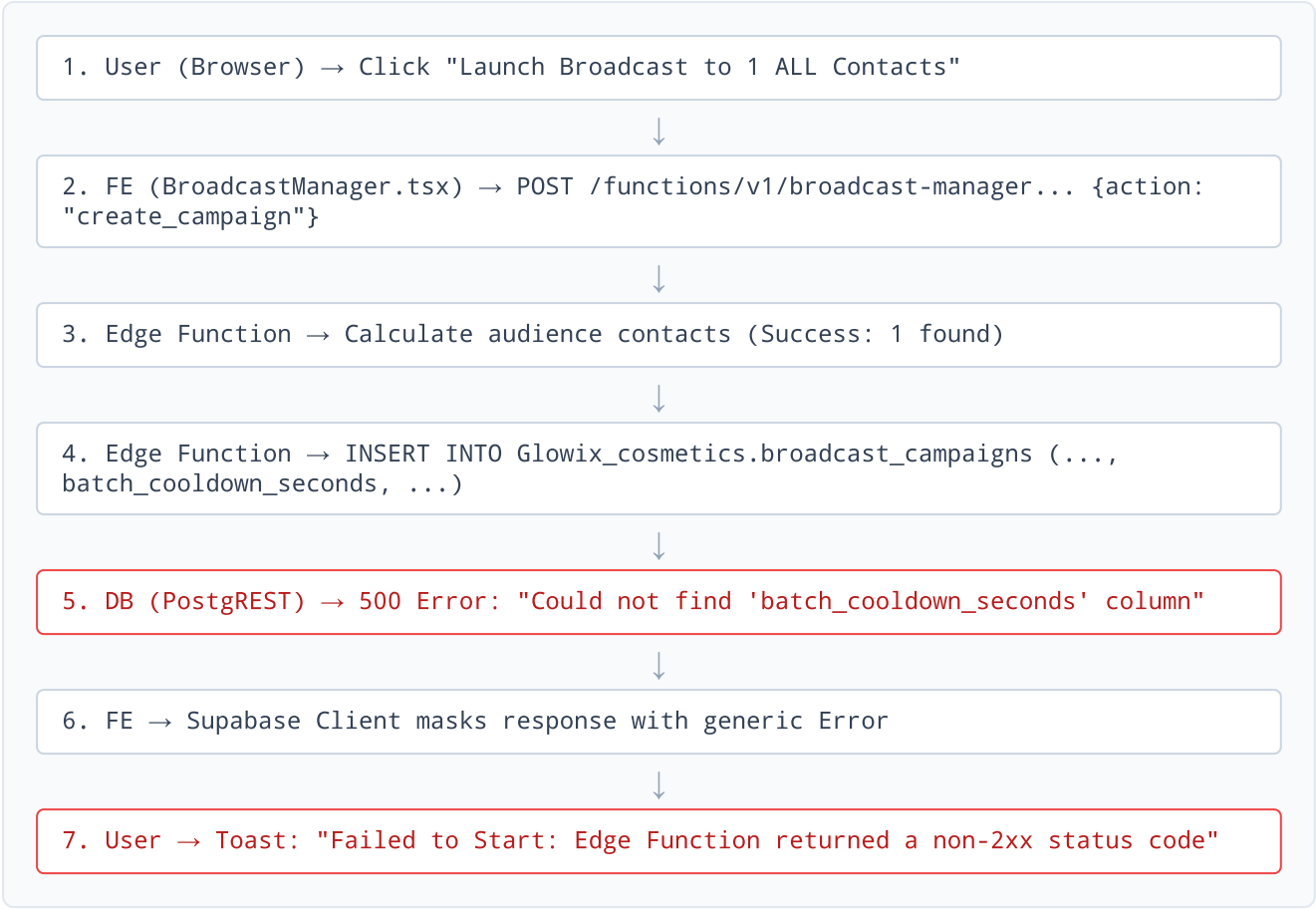
2. Technical Root Cause Breakdown

A. The Evolution of the Schema

In the repository, the table was introduced across two separate milestones:

- Initial Baseline (db/01_schema.sql):** Defined the table with only basic fields (id, user_id, title, message, total_count, status, delay_seconds, etc.).
- Commit d98613e (Advanced Broadcast & Anti-Ban Controls):** Added rate-limiting and anti-ban configuration options including name, segment, message_template, media_type, total_recipients, delay_seconds_min, delay_seconds_max, **batch_size**, **batch_cooldown_seconds**, started_at, and completed_at.

C. Execution Sequence Flow



3. Schema Difference Matrix

The following table maps the discrepancies between the live database and the expected application state.

Column Name	In 01_schema.sql (Live DB)	In Edge Function / New Code	Missing in Live DB?
id / user_id / title	Yes	Yes	No
name	No	Yes	YES
segment	No	Yes	YES
message	Yes	Yes	No
message_template	No	Yes	YES
media_url	Yes	Yes	No
media_type	No	Yes	YES
total_count	Yes	Yes	No
total_recipients	No	Yes	YES
sent_count / failed_count	Yes	Yes	No
delay_seconds	Yes	Yes	No
delay_seconds_min	No	Yes	YES

4. Step-by-Step Fix Instructions

Step 1: Run Idempotent Schema Migration in Supabase

Log into your Supabase Dashboard at <https://supabase.buildstart.io>, navigate to the **SQL Editor**, copy and paste the following SQL script, and click **Run**.

```
-- ===== -- 1. Ensure
Table Exists -- ===== CREATE
TABLE IF NOT EXISTS Glowix_cosmetics.broadcast_campaigns ( id uuid DEFAULT gen_random_uuid()
NOT NULL PRIMARY KEY, user_id uuid NOT NULL, name text DEFAULT 'Promotional Broadcast'::text
NOT NULL, title text DEFAULT 'Promotional Broadcast'::text NOT NULL, segment text DEFAULT
'all'::text NOT NULL, audience_filter text DEFAULT 'all'::text NOT NULL, message text,
message_template text, media_url text, media_type text, total_recipients integer DEFAULT 0
NOT NULL, total_count integer DEFAULT 0 NOT NULL, sent_count integer DEFAULT 0 NOT NULL,
failed_count integer DEFAULT 0 NOT NULL, status text DEFAULT 'draft'::text NOT NULL,
delay_seconds integer DEFAULT 10 NOT NULL, delay_seconds_min integer DEFAULT 8 NOT NULL,
delay_seconds_max integer DEFAULT 15 NOT NULL, batch_size integer DEFAULT 30 NOT NULL,
batch_cooldown_seconds integer DEFAULT 120 NOT NULL, started_at timestamp with time zone,
completed_at timestamp with time zone, created_at timestamp with time zone DEFAULT now() NOT
NULL, updated_at timestamp with time zone DEFAULT now() NOT NULL ); --
===== -- 2. Explicitly ADD all
missing columns (idempotent for existing DBs) --
===== ALTER TABLE
Glowix_cosmetics.broadcast_campaigns ADD COLUMN IF NOT EXISTS name text DEFAULT 'Promotional
Broadcast'::text NOT NULL; ALTER TABLE Glowix_cosmetics.broadcast_campaigns ADD COLUMN IF NOT
EXISTS title text DEFAULT 'Promotional Broadcast'::text NOT NULL; ALTER TABLE
Glowix_cosmetics.broadcast_campaigns ADD COLUMN IF NOT EXISTS segment text DEFAULT
'all'::text NOT NULL; ALTER TABLE Glowix_cosmetics.broadcast_campaigns ADD COLUMN IF NOT
EXISTS audience_filter text DEFAULT 'all'::text NOT NULL; ALTER TABLE
Glowix_cosmetics.broadcast_campaigns ADD COLUMN IF NOT EXISTS message text; ALTER TABLE
Glowix_cosmetics.broadcast_campaigns ADD COLUMN IF NOT EXISTS message_template text;
```

```

ALTER TABLE Glowix_cosmetics.broadcast_campaigns ADD COLUMN IF NOT EXISTS media_url
text; ALTER TABLE Glowix_cosmetics.broadcast_campaigns ADD COLUMN IF NOT EXISTS media_type
text; ALTER TABLE Glowix_cosmetics.broadcast_campaigns ADD COLUMN IF NOT EXISTS
total_recipients integer DEFAULT 0 NOT NULL; ALTER TABLE Glowix_cosmetics.broadcast_campaigns
ADD COLUMN IF NOT EXISTS total_count integer DEFAULT 0 NOT NULL; ALTER TABLE
Glowix_cosmetics.broadcast_campaigns ADD COLUMN IF NOT EXISTS sent_count integer DEFAULT 0
NOT NULL; ALTER TABLE Glowix_cosmetics.broadcast_campaigns ADD COLUMN IF NOT EXISTS
failed_count integer DEFAULT 0 NOT NULL; ALTER TABLE Glowix_cosmetics.broadcast_campaigns ADD
COLUMN IF NOT EXISTS delay_seconds_min integer DEFAULT 8 NOT NULL; ALTER TABLE
Glowix_cosmetics.broadcast_campaigns ADD COLUMN IF NOT EXISTS delay_seconds_max integer
DEFAULT 15 NOT NULL; ALTER TABLE Glowix_cosmetics.broadcast_campaigns ADD COLUMN IF NOT
EXISTS batch_size integer DEFAULT 30 NOT NULL; ALTER TABLE
Glowix_cosmetics.broadcast_campaigns ADD COLUMN IF NOT EXISTS batch_cooldown_seconds integer
DEFAULT 120 NOT NULL; ALTER TABLE Glowix_cosmetics.broadcast_campaigns ADD COLUMN IF NOT
EXISTS started_at timestamp with time zone; ALTER TABLE Glowix_cosmetics.broadcast_campaigns
ADD COLUMN IF NOT EXISTS completed_at timestamp with time zone; --
===== -- 3. Update Status Check
Constraint -- ===== ALTER TABLE
Glowix_cosmetics.broadcast_campaigns DROP CONSTRAINT IF EXISTS
broadcast_campaigns_status_check; ALTER TABLE Glowix_cosmetics.broadcast_campaigns ADD
CONSTRAINT broadcast_campaigns_status_check CHECK ((status = ANY (ARRAY['draft'::text,
'sending'::text, 'in_progress'::text, 'completed'::text, 'paused'::text, 'cancelled'::text,
'failed'::text]))); -- ===== --
4. Queue Table & Columns --
===== CREATE TABLE IF NOT
EXISTS Glowix_cosmetics.broadcast_queue ( id uuid DEFAULT gen_random_uuid() NOT NULL PRIMARY
KEY, campaign_id uuid NOT NULL REFERENCES Glowix_cosmetics.broadcast_campaigns(id) ON DELETE
CASCADE, user_id uuid NOT NULL, phone_number text NOT NULL, customer_name text,
recipient_name text, status text DEFAULT 'pending'::text NOT NULL, error_message text,
retry_count integer DEFAULT 0 NOT NULL, sent_at timestamp with time zone, created_at
timestamp with time zone DEFAULT now() NOT NULL ); ALTER TABLE
Glowix_cosmetics.broadcast_queue ADD COLUMN IF NOT EXISTS customer_name text; ALTER TABLE
Glowix_cosmetics.broadcast_queue ADD COLUMN IF NOT EXISTS recipient_name text; ALTER TABLE
Glowix_cosmetics.broadcast_queue ADD COLUMN IF NOT EXISTS error_message text; ALTER TABLE
Glowix_cosmetics.broadcast_queue ADD COLUMN IF NOT EXISTS retry_count integer DEFAULT 0 NOT
NULL; ALTER TABLE Glowix_cosmetics.broadcast_queue ADD COLUMN IF NOT EXISTS sent_at timestamp
with time zone;

```

```

ALTER TABLE Glowix_cosmetics.broadcast_queue DROP CONSTRAINT IF EXISTS
broadcast_queue_status_check; ALTER TABLE Glowix_cosmetics.broadcast_queue ADD CONSTRAINT
broadcast_queue_status_check CHECK ((status = ANY (ARRAY['pending'::text, 'sending'::text,
'sent'::text, 'failed'::text]))); --
===== -- 5. Indexes, Row Level
Security & Grants -- =====
CREATE INDEX IF NOT EXISTS idx_broadcast_campaigns_user_id ON
Glowix_cosmetics.broadcast_campaigns USING btree (user_id); CREATE INDEX IF NOT EXISTS
idx_broadcast_queue_campaign_id ON Glowix_cosmetics.broadcast_queue USING btree
(campaign_id); ALTER TABLE Glowix_cosmetics.broadcast_campaigns ENABLE ROW LEVEL SECURITY;
ALTER TABLE Glowix_cosmetics.broadcast_queue ENABLE ROW LEVEL SECURITY; DROP POLICY IF EXISTS
"Users can manage own broadcast campaigns" ON Glowix_cosmetics.broadcast_campaigns; CREATE
POLICY "Users can manage own broadcast campaigns" ON Glowix_cosmetics.broadcast_campaigns FOR
ALL USING ((auth.uid() = user_id)) WITH CHECK ((auth.uid() = user_id)); GRANT ALL ON TABLE
Glowix_cosmetics.broadcast_campaigns TO anon, authenticated, service_role; GRANT ALL ON TABLE
Glowix_cosmetics.broadcast_queue TO anon, authenticated, service_role; --
===== -- 6. Force PostgREST to
reload its schema cache --
===== NOTIFY pgrst, 'reload
schema';

```

Step 2: Verification

1. Return to the browser tab (`glowix-cosmetics-customization.vercel.app/dashboard/settings`).
2. Refresh the page (Ctrl + Shift + R).
3. Click "**Launch Broadcast to 1 ALL Contacts**".
4. The request will now return **HTTP 200 OK**, create the campaign in draft/sending state, populate the recipient queue, and begin the safe dispatch loop.

5. Recommended Codebase Improvements

Improve UI Error Reporting

Currently, in `BroadcastManager.tsx`, errors are displayed generically via Supabase. To ensure future schema or validation issues immediately show actionable messages in the UI, parse runtime errors from `error.context`:

```

async function extractInvokeError(error: any): Promise<string> { if (!error) return
"Unknown error"; try { if (error.context?.json) { const body = await error.context.json();
return body?.error || body?.message || error.message; } } catch (_) {} return error.message;
}

```